

EECS 482

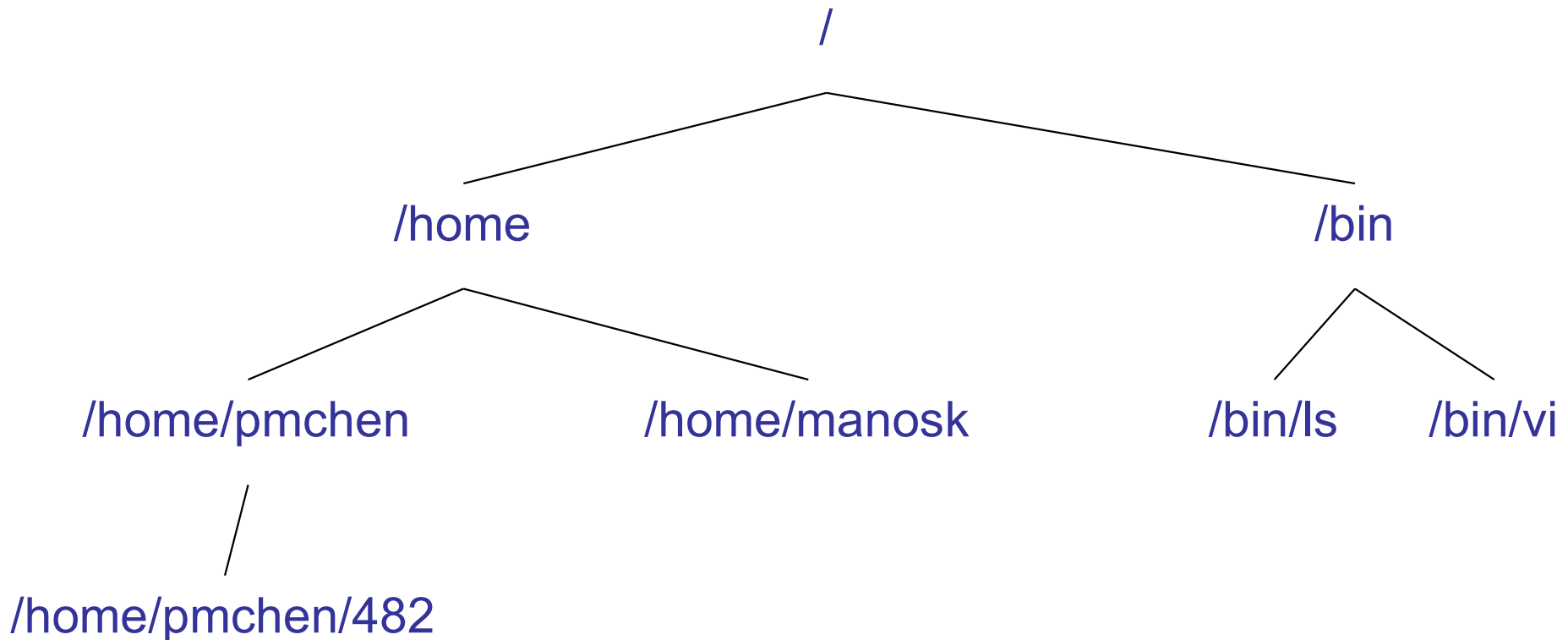
Introduction to operating systems

Hierarchical file systems; caching

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Structured naming

- How does a file system organize files?



Tree of directories

- What is needed for a tree?
 - Nodes (with one node designated as the root node)
 - Pointers between nodes
 - Exactly one path from root node to each other node
- Hierarchical file system
 - Nodes: directories, files
 - Pointers between nodes: directory entries
- Directory entries
 - File name → disk block # for header
 - Directory name → disk block # for header
 - Another way to treat directories like files

Support for structured naming

- How to map from a hierarchically-structured file name (e.g., /home/pmchen/482/notes) to the disk block # of the file header?

How to read
</home/pmchen/482/notes> ?

Unified view of multiple storage devices

- Combine multiple storage devices into a file system
 - Each device contains own file system (starting with its root)
 - A directory entry can point to the root of a **different device**
- Example: login.engin.umich.edu
 - / (root)
 - bin (same device as /)
 - tmp (file system on separate storage device)
 - afs (file system on network servers)
- Directory entries
 - File
 - Directory
 - Device

File caching

- Lots of on-disk data
 - File data
 - Directories
 - Inodes, etc.
- How to improve performance
 - Random I/O → sequential I/O
 - Disk I/O → memory accesses
 - » What single number would you cache to accelerate reading files in /home/pmchen/482?

File caching

- Should cache be in physical or virtual memory?
- Should cache be write-through or write-back?



- Project 3?

Comparing virtual memory and file caching

Address spaces

Memory

Disk

Increase capacity via
swap space

File systems

Memory

Disk

Increase speed via
file cache

- Both use physical memory as a cache for disk
 - **Virtual memory**: Use disk for increased capacity
 - **File systems**: Use memory for faster performance
- Both compete for physical memory
 - Local vs. global replacement
- **Why have two separate mechanisms for caching disk data in memory?**

Memory-mapped files

- Use the VM paging system to cache both virtual address space **and** file system data
 - Map file into a virtual address space
 - Point the backing store for that part of the address space at the file's data blocks
- Example: how to load a program executable from disk to memory
 - `vm_map` the executable file into the address space
 - Load the pages from the file on demand as the process executes

Memory map of Linux process

Multiple updates and reliability

- File system must ensure reliability/durability
 - Data must be consistent, even in the presence of crashes and power outages
- Operations that require multiple updates make this challenging
 - Transfer \$100 from Peter's account to Janet's account
 1. Deduct \$100 from Peter's account
 2. Add \$100 to Janet's account
 - Move file from old directory to new directory
 1. Delete file from old directory
 2. Add file to new directory

Multiple updates and reliability

What disk I/Os are needed to create a new file?

- Option 1:
 1. Update directory to point to new file header
 2. Write new file header

What happens if crash occurs between #1 and #2?
- Option 2:
 1. Write new file header
 2. Update directory to point to new file header

What happens if crash occurs between #1 and #2?

Does the file exist at this point?

Is careful ordering enough?

- Example: Create new file and update free block list
- Option 1:

1. Write new file header
2. Update directory to point to new file header
3. Write the new free block list

What happens if crash occurs between #2 and #3?

- Option 2:

1. Write new file header
2. Write the new free block list
3. Update directory to point to new file header

What happens if crash occurs between #2 and #3?

Is careful ordering enough?

- Transfer \$100 from Peter's account to Janet's account
- Option 1:
 1. Deduct \$100 from Peter's account
 2. Add \$100 to Janet's account

What happens if crash occurs between #1 and #2?
- Option 2:
 1. Add \$100 to Janet's account
 2. Deduct \$100 from Peter's account

What happens if crash occurs between #1 and #2?

What does this remind you of?
